Jev-Backed Function Calling: Iterative Repair with Cumulative Decision Feedback

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Abstract

Function-calling models must choose a function and input values that match the user's request. Correct format alone does not ensure correct choices. We present Jev-Backed Function Calling (**JBFC**), an inference-time method that uses TypeSafe's System One (Decision) model *Jev* to check an LLM's proposed call. *Jev* checks function choice and, when a call is present, its arguments with binary questions. If a required check fails, we add the proposal and judgments to a growing history placed before the original user request. The LLM uses this feedback to revise both the function and arguments. Acceptance requires all applicable binary checks and the schema check to pass on the same proposal. A repair limit is chosen on development data and fixed before testing. Our code compares seven methods on 2,748 test examples from a 3,051-example Berkeley Function Calling Leaderboard (BFCL) subset, measuring call correctness, function choice, argument correctness, time, and cost. Experiments have not yet been run. They will test whether accumulated verification feedback improves calls and whether repairs justify their added cost.

1 Introduction

To carry out a user's request correctly, a function-calling system must choose the right function and provide the right argument values. A language model, called the generator, makes these choices. However, a call can have the required format and still use the wrong function or values. For example, a travel assistant may choose the wrong booking function or swap the departure and destination. A JSON schema check verifies required inputs and their types, but cannot guarantee that the call matches the user's request.

Tool-use training and evaluation have improved function-calling systems [2, 3, 1]. Even so, the generated function and arguments can still be wrong. We study whether checks from a separate decision model can guide the LLM to improve its next proposal before execution.

JBFC uses *Jev*, TypeSafe's System One model, to answer binary questions about an LLM's function choice and arguments [12]. When a check fails, we keep the proposed call and its judgments and place all earlier records before the original user request. The LLM then proposes a new call using this accumulated feedback. It can change both the function and argument values. **JBFC** accepts a proposal only when all applicable binary checks and the schema check pass on that proposal.

Earlier work already uses checks and feedback to fix calls. ToolVerifier checks tool choice and parameters [9]; ToolCritic gives feedback on tool-use errors [10]. ReAct uses reasoning, actions, and observations to guide later steps [16]. **JBFC** also carries feedback forward in the prompt, but its feedback comes from checking proposed calls before execution. Our contribution is this specific use of two *Jev* judgments and accumulated prompts, together with an evaluation of its effects and costs.

We compare **JBFC** with direct calling, schema repair, self-revision, single-check versions, and a latest-feedback version. A repair limit chosen on development data controls the comparison. We measure errors fixed or introduced by repair, and keep call correctness separate from acceptance, time, and cost.

2 Related Work

Tool use and benchmarks. Gorilla studies API-call generation [2]; ToolLLM studies tool-use training and evaluation [3]. BFCL tests function calling across many settings [1]. We use its single-turn Python categories to study decisions made before a function runs. API-Bank offers dialogues with executable API calls [4]. StableToolBench uses caching and simulation to handle changing APIs [5]. These suggest broader tests beyond our BFCL subset.

Checks and feedback. Self-Refine uses self-generated feedback [6], Reflexion uses written feedback [7], and CRITIC uses feedback from tools [8]. Self-correction can turn correct answers into wrong ones [11], so we measure both fixes and new errors. ToolVerifier compares alternatives to check tool choice and parameters separately [9]. ToolCritic identifies tool-use errors and returns feedback [10]. **JBFC** checks complete proposals with a fixed external model through an API and limits repairs.

Accumulated prompts. ReAct combines reasoning with actions and observations from an environment [16]. Our method uses a similar idea of carrying earlier feedback into the next prompt. Each record contains a proposed function call and verification results, rather than an execution result. The LLM receives all records before the unchanged user request. We do not require a generated reasoning trace or run functions during repair.

***Jev* as a judge.** Li et al.'s *JEV-as-a-Judge* [15] checks finished outputs with probabilities for defined verdicts. If a verdict is uncertain, an independent, stronger judge decides. With fixed rules for when to use the stronger judge, it lowers

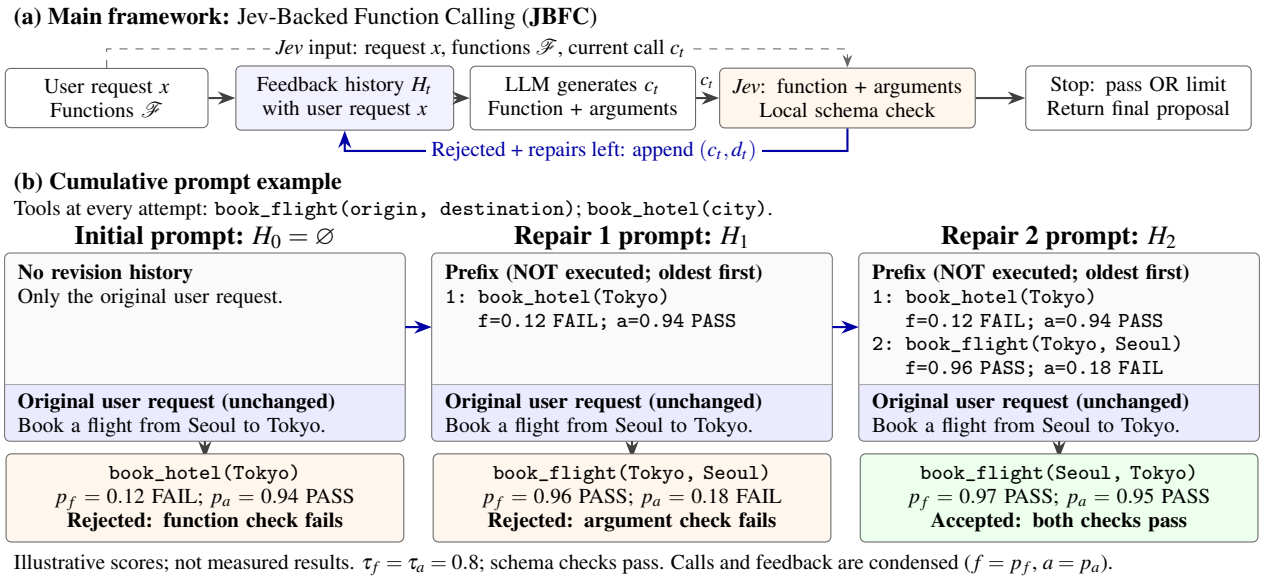


Figure 1: **JBFC** and cumulative prompt feedback. Top: the LLM reads earlier calls and judgments (H_t), then the original request (x), and generates a call (c_t) with a function and arguments. Failed checks add a record for another attempt. Bottom: growing history precedes the unchanged request; new calls appear below. Argument checks concern the proposed function. No function runs during repair.

judging cost while keeping similar accuracy on some tasks. It has limits on difficult reasoning, misleading answer styles, and writing judged without reference information. The study does not test tool or action selection inside agents. **JBFC** checks proposed function calls and returns failed checks to the same generator for revision. We test the specific design of **JBFC**.

3 Method

3.1 Propose and Check a Call

Let x be the user request, $\mathscr{F}$ the available functions, and G the generating LLM. A proposal is $c_t = (f_t, a_t)$, with function f_t and arguments a_t, or $c_t = \varnothing$ for no call. We study single-call-or-no-call decisions.[1] Local code checks readability, function availability, required arguments, and types. Write $S(c_t) = 1$ when these checks pass.

Binary verification: JBFC. For each readable proposal, one *Jev* request checks the function choice and, when a call is present, its arguments:

$$c_t = G(H_t, x, \mathscr{F}), \tag{1}$$

$$p_f^{(t)} = J_f(x, \mathscr{F}, c_t), \qquad p_a^{(t)} = J_a(x, \mathscr{F}, c_t). \tag{2}$$

Here H_t contains earlier proposals and feedback and is initially empty. The function `Noul` question asks whether the function choice, including no call, fits the request. The argument question checks required values, their support in the request, and their parameter roles. For no call, it is skipped and $p_a^{(t)}$ is undefined. A selected function with missing arguments still receives both checks. Both must pass for a call.

[1] We treat single-turn decisions as the basic unit of function calling. Parallel calling combines multiple such decisions, while multi-turn interactions repeat them with updated context. We therefore evaluate the verification and repair loop on this basic unit and exclude parallel and multi-turn settings from this study.

3.2 Accept or Repair a Call

Acceptance requires the schema check and all required *Jev* checks to pass on the same proposal. The acceptance rule is

$$A_t = \begin{cases} S(c_t)\,\mathbf{1}[p_f^{(t)} \geq \tau_f], & c_t = \varnothing, \\ S(c_t)\,\mathbf{1}[p_f^{(t)} \geq \tau_f]\,\mathbf{1}[p_a^{(t)} \geq \tau_a], & \text{otherwise.} \end{cases} \tag{3}$$

`Noul` returns the probability of yes, with no separate confidence field [13, 14]. We start with $\tau_f = \tau_a = 0.8$, untuned defaults chosen for this experiment.

Feedback before the user request. H_t stores earlier records. The current record (c_t, d_t) pairs proposal c_t with feedback d_t (probabilities, pass/fail flags, thresholds, and local errors). No-call skips the argument check. After rejection, if repairs remain, $\|$ adds $[(c_t, d_t)]$, a list containing this one record, to the end of H_t:

$$H_{t+1} = H_t \,\|\, [(c_t, d_t)]. \tag{4}$$

The LLM reads this history before the unchanged request and can change both function and arguments. Schemas remain available; system instructions stay separate.

Algorithm 1 rechecks up to $r+1$ proposals: the first plus repairs (Python default $r = 3$; development-tuned in experiments). Its `Prefix` function places JSON records, oldest first, before unchanged original messages. Empty history adds no

Algorithm 1 JBFC: verification and prompt repair

Require: Request x, functions $\mathscr{F}$, generator G, *Jev* J, repair limit r, thresholds τ_f, τ_a

$H \leftarrow [\,]$
for $t = 0, \ldots, r$ **do**
  $c \leftarrow G(\text{Prefix}(H, x), \mathscr{F})$
  $(s, e, \text{readable}) \leftarrow \text{SchemaCheck}(c, \mathscr{F})$
  $p_f, p_a \leftarrow \text{null}, \text{null}$
  if readable **then**
    $Q \leftarrow \{\text{function}\}$
    if $c \neq \varnothing$ **then** $Q \leftarrow Q \cup \{\text{arguments}\}$
    $(p_f, p_a) \leftarrow J(x, \mathscr{F}, c, Q)$
  end if
  $A \leftarrow \text{Accept}(s, c, p_f, p_a, \tau_f, \tau_a)$ ▷ Eq. 3
  $d \leftarrow \text{Feedback}(e, p_f, p_a, \tau_f, \tau_a, c)$
  if $A = 1$ **or** $t = r$ **then break**
  $H \leftarrow H \,\|\, [(c, d)]$
end for
return c

prefix. Records include round numbers and mark proposals as not executed. Here $s = S(c)$ and e lists local errors. Readable proposals receive *Jev* checks even if their schema fails; unreadable ones receive local feedback. *Jev* never receives history. Unasked p_a stays null. Null required probabilities cannot pass the gate; invalid API responses stop the run. Reference answers stay out of model requests. API errors stop the run after transport retries, separate from repairs.

3.3 Final Proposal

When a proposal passes all applicable checks, we return it. If the repair limit is reached first, we return the last proposal even if it fails a check (`execute_last`). We score the final proposal against the benchmark answers without running the function.

4 Experiments

4.1 Data and Evaluation

We prepare 3,051 BFCL examples from seven categories at repository revision `6ea57973c7a6`[2]. Table 1 shows downloaded and test counts. We sort each category by the SHA-256 hash of `seed:id`. The first $\max(1, \lfloor 0.1 N_k \rfloor)$ examples form the development set; N_k is the category size. Seed 1729 gives 303 development and 2,748 test examples. Model experiments have not been run.

For categories with reference calls, the original Python abstract syntax tree (AST) checker tests function names and arguments against allowed answers. Irrelevance categories require no call. Live relevance checks only call presence, not function or argument correctness. We exclude it from main accuracy and from Brier scores, which need those labels. The main score covers 2,733 test examples per repeated run. We report category accuracy and a micro-average, which weights examples equally. Category scores show what the larger live categories may hide in the average.

[2]Full revision: `6ea57973c7a6097fd7c5915698c54c17c5b1b6c8`. The saved setup file lists source URLs, file hashes, and format changes; the hashed split files list the example IDs.

Category	All	Test	Direct	**JBFC**
Simple Python	400	360	–	–
Multiple	200	180	–	–
Irrelevance	240	216	–	–
Live simple	258	233	–	–
Live multiple	1053	948	–	–
Live irrelevance	884	796	–	–
Live relevance[†]	16	15	–	–
Total	3051	2748	–	–

Table 1: BFCL example counts and test accuracy (%). Counts describe the prepared data. [†]Live relevance checks only whether a call is present; total accuracy excludes these 15 cases. – means the experiment has not been run.

BFCL's "multiple" categories offer several functions to choose from, without parallel calls. We exclude parallel calls, multi-turn state, web search, memory, and live execution. "Live" is a category name. AST checks compare call structure with annotated alternatives; passing them does not prove execution success, authorization, or task completion.

4.2 Models and Methods

The generator is `gpt-5.4-mini-2026-03-17`, with reasoning effort `none`; *Jev* is pinned to `jev-1.13.0`. We use OpenAI Responses and System One through an APIM gateway, convert BFCL schemas to JSON Schema with API-safe function names, and plan three repeated runs with fixed prompts. Models, thresholds, and prompts can be changed, but their selection must use development data before the test setup is frozen.

We compare seven methods, sharing one initial proposal per example and run. *Direct* uses the initial proposal; *Schema-only* repairs schema errors. *Self-refine* uses generic self-review in all r repairs, not the full Self-Refine method [6]. **JBFC-F** and **JBFC-A** obtain applicable `Noul` judgments but expose and require only the function or argument verdict, plus schema checks. **JBFC-A** accepts no call on schema checks alone. **JBFC** requires all applicable checks. **JBFC-L** shares its settings and initial proposal but shows only the latest rejected proposal and feedback. Other repair methods show all earlier records before the request. Costs can differ.

Choosing the repair limit. We choose r before testing. The initial candidates are $\mathscr{R} = \{0, 1, 2, 3, 4, 5\}$, a configurable search range. **JBFC** has no built-in limit of five. In three development runs, we allow up to $\max \mathscr{R}$ repairs and stop early when all applicable checks pass. Neither model sees the limit, and no function runs during repair. Each smaller limit can therefore be tested on the same saved sequence without more API calls. *Jev* thresholds stay at 0.8 during this search.

We choose the limit with the highest final-call accuracy on development data. Ties choose the smaller limit:

$$r^* = \min \arg\max_{r \in \mathscr{R}} \widehat{\text{Acc}}_{\text{dev}}(r). \tag{5}$$

The score covers 302 development examples per run, excluding the one live-relevance example. We save the cho-

Method	Call	Func.	Args.	Sec.	USD/1k
Direct	–	–	–	–	–
Schema repair	–	–	–	–	–
Self-refine	–	–	–	–	–
JBFC-F	–	–	–	–	–
JBFC-A	–	–	–	–	–
JBFC-L	–	–	–	–	–
JBFC	–	–	–	–	–

Table 2: Methods on the BFCL test set. The repair limit is chosen on development data. Call and function accuracy (%) exclude live relevance. Argument accuracy (%) uses calls with the correct function and argument labels. – means unavailable or unrun.

r	Dev Acc. (%)	Repairs	Sec.	USD/1k
0	–	–	–	–
1	–	–	–	–
2	–	–	–	–
3	–	–	–	–
4	–	–	–	–
5	–	–	–	–

Table 3: Development data choose the repair limit for **JBFC**. We choose the highest call accuracy; ties choose the smaller limit. Accuracy excludes live relevance. – means no result is available or the experiment has not been run.

sen limit, settings, and development-record hashes before three test runs compare all seven methods. Test scores never choose the limit. Table 3 shows the development search for **JBFC**. Optional smaller-limit analyses use only saved **JBFC** sequences. Cost and time cover only the selected prefix. With $r = 0$, the initial proposal is checked but never revised.

4.3 Measures, Cost, and Uncertainty

We report final call accuracy, function accuracy, and argument accuracy conditional on the correct function. Every score includes its denominator. Live relevance is excluded from these accuracy measures and reported separately. We compare initial and final candidates to identify function changes, argument fixes, and new errors after feedback.

Acceptance is distinct from correctness. Coverage is the fraction accepted; accepted precision is accuracy among accepted calls. Selective risk is one minus this precision. False acceptance conditions on incorrect calls, and false rejection on correct calls. We also record corrections, regressions, repairs, API errors, and Brier scores where matching probability and correctness labels exist. These measures show whether feedback corrects rejected errors or instead introduces new ones.

Reported costs count all recorded generation and verification work, with shared initial calls charged to each method. Growing history adds input tokens on later attempts, and these tokens count toward cost. We report mean and p95 API time without retry waits, token use, estimated cost, and per-run scores. Test costs exclude the development search. Missing usage or prices remain unavailable, not zero. Charges for failed requests may be missing. Default prices on October 15, 2026, per million tokens are \$0.75/\$0.075/\$4.50 for uncached/cached/output LLM tokens and \$0.042 for *Jev* input, with free *Jev* output; gateway fees are excluded.

We compare repairs with Direct and **JBFC** with **JBFC-L**, matching examples, runs, and initial proposals. We average accuracy differences across runs per example, then bootstrap IDs 2,000 times with seed 1729 for 95% intervals. Histories differ from repair 2; $r \leq 1$ cannot test cumulative history. Tables 2–3 remain unrun; partial or synthetic runs cannot fill them.

4.4 Study Limits

The verifier can accept a wrong call or share the generator's mistake. Repair can introduce errors, and later proposals may differ from development examples. Accumulated feedback can reinforce an earlier mistake or distract the LLM from the original request. The history also grows with each repair, increasing context use and cost. Binary verdicts do not explain how to fix an error. Higher acceptance alone does not show improvement.

The study uses one model pair, mostly English requests, and single-turn calls. Public examples may overlap with training data. We keep ten documented annotation inconsistencies in the main score. In six, reference values contradict declared types, so schema-checked methods cannot accept a correct call. The candidate grid and small development set can affect the chosen repair limit. A common limit chosen for **JBFC** may disadvantage other methods. Further studies should tune each method separately and test more models and domains. API failures and incomplete runs remain visible. Conclusions must follow measured results, including no gain or worse performance.

5 Conclusion

JBFC uses *Jev*'s function and argument judgments as accumulated prompt feedback. After a failed check, earlier proposals and judgments appear before the original user request, allowing the LLM to revise the complete call. A development-chosen repair limit bounds this process. The code separates correctness, acceptance, and cost. Experiments remain to be run; they will test whether this feedback improves function calling and whether the gains justify the added work.

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